



9/1/2026

N + - Network Fundamentals & Building Networks.

Switching and VLANS.

TASK 1.

- ❖ Draw a simple network diagram showing three devices (like your laptop, a friend's phone, and a smart TV) connected to a switch, and label which device is in which VLAN (e.g., VLAN 10 for streaming, VLAN 20 for personal devices).

➤ VLAN Network Diagram:

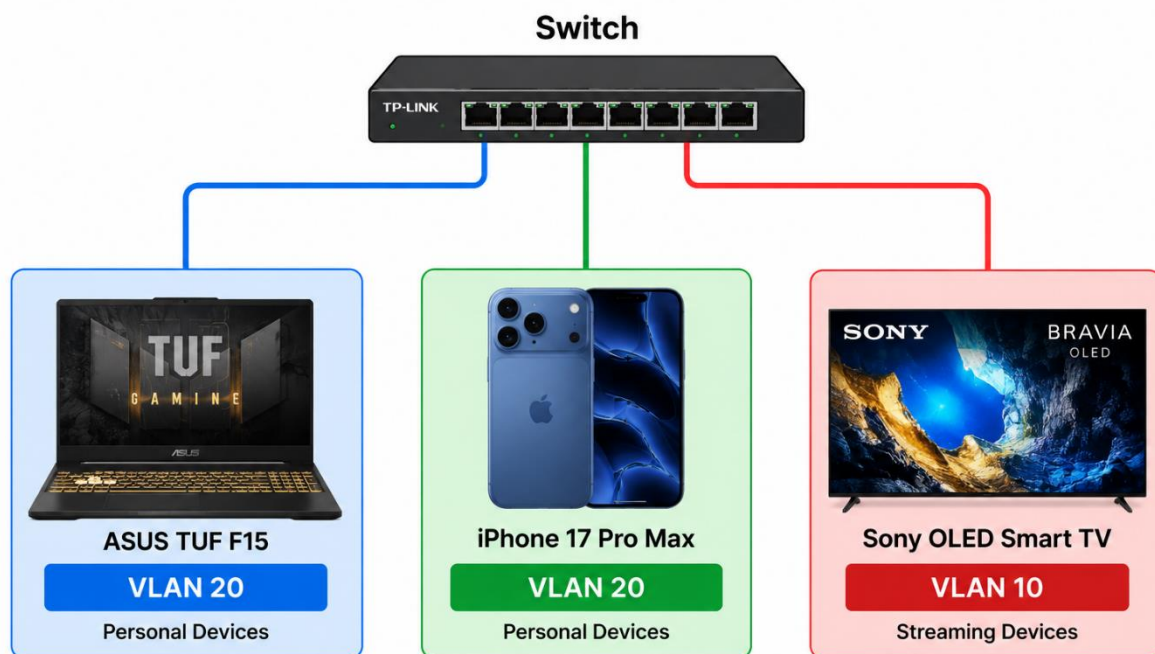


Figure : Network Diagram Showing Devices Connected to a Switch and Their VLANs.

- ⌘ **ASUS TUF F15 → VLAN 20: Personal Devices**
- ⌘ **iPhone 17 Pro Max → VLAN 20: Personal Devices**
- ⌘ **Sony OLED Smart TV → VLAN 10: Streaming Devices**

TASK 2.

- ❖ **Configure a simulated switch using Cisco Packet Tracer or NetSim: create two VLANs (VLAN 10 and VLAN 20), assign three PCs to VLAN 10 and two PCs to VLAN 20, and verify that PCs in different VLANs cannot ping each other, Use the 'vlan' and 'switchport access vlan' commands in the simulator CLI.**

➤ **VLAN Configuration and Verification:**

1. Create the VLANs

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan 10
Switch(config-vlan)# name VLAN10
Switch(config-vlan)# exit
Switch(config)# vlan 20
Switch(config-vlan)# name VLAN20
Switch(config-vlan)# exit
```

2. Assign PCs to VLANs

Assign **PC1, PC2** and **PC3** to VLAN 10:

```
Switch(config)# interface range fa0/1-3
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)# exit
```

Assign **PC4** and **PC5** to VLAN 20:

```
Switch(config)# interface range fa0/4-5
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
Switch(config-if-range)# exit
```

3. Verify VLAN Configuration

Switch# show vlan brief

The output should show **PC1–PC3 in VLAN 10** and **PC4–PC5 in VLAN 20**.

4. Verify Communication

When a PC from **VLAN 10** tries to ping a PC from **VLAN 20**, the ping should fail because devices in different VLANs are logically separated.

Result: PCs within the same VLAN can communicate, while PCs in different VLANs cannot communicate without inter-VLAN routing.

TASK 3.

❖ In any network simulator, set up a scenario where you move a PC from VLAN 10 to VLAN 20. Show the before and after configuration commands and test connectivity to confirm the change.

➤ Moving a PC from VLAN 10 to VLAN 20:

1. Before Configuration — PC1 in VLAN 10

Switch> enable

Switch# configure terminal

Switch(config)# interface fa0/1

Switch(config-if)# switchport mode access

Switch(config-if)# switchport access vlan 10

Switch(config-if)# exit

⌘ Connectivity Test:

PC1> ping 192.168.20.2

Result: Ping fails because PC1 is in **VLAN 10**.

2. After Configuration — PC1 moved to VLAN 20

Switch(config)# interface fa0/1

Switch(config-if)# switchport access vlan 20

Switch(config-if)# exit

⌘ Connectivity Test:

PC1> ping 192.168.20.2

Result: Ping succeeds if the destination PC is also in **VLAN 20**.

Change Confirmed: PC1 has been successfully moved from **VLAN 10** to **VLAN 20**.

TASK 4.

❖ Write a short script or command list that would automate assigning 10 ports on a switch to VLAN 30, using the syntax of your chosen simulator (Packet Tracer, NetSim, or GNS3), Use a loop or range command if supported by the tool to avoid repeating the same command 10 times.

➤ Assigning 10 Ports to VLAN 30:

Simulator Used: Cisco Packet Tracer

Switch> enable

Switch# configure terminal

Switch(config)# vlan 30

```
Switch(config-vlan)# exit
Switch(config)# interface range fa0/1-10
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 30
Switch(config-if-range)# exit
Switch(config)# end
```

This interface range fa0/1-10 command assigns 10 ports at once to VLAN 30, avoiding repetition.